

Zoeb Ali Khan

M.Sc. Data Science Student · Universität Bielefeld

Bielefeld, Germany · +49 1551 0592355 · zoebalikhan00@gmail.com · GitHub · LinkedIn

PROFILE

M.Sc. Data Science student at Universität Bielefeld with practical experience in machine learning research, deep learning experimentation, and data engineering for real-world datasets. Genuine interest in climate tech and renewable energy backed by prior project work on German solar and wind generation data. Comfortable working with PyTorch, TensorFlow, and Scikit-learn; experienced taking ML prototypes from concept to deployed application. Available from August 2026.

EDUCATION

M.Sc. Data Science - Universität Bielefeld, Germany

Ongoing, expected March 2027

Coursework: Machine Learning, NLP, Statistical Modeling, Data Mining, Probabilistic Models, Data Engineering

B.E. Artificial Intelligence & Data Science - University of Mumbai, India

08/2020 – 06/2024

EXPERIENCE

AI & Machine Learning Intern - YBI Foundation

05/2024 – 07/2024

Remote / India

- Developed and deployed a supervised ML classification model for credit-risk prediction, handling feature engineering, model selection, and evaluation end to end.
- Built Python and SQL ETL pipelines into Google Cloud Platform (BigQuery) and automated CRM reporting workflows, reducing manual effort by an estimated 5+ hours per week with validated, audit-ready outputs.

PROJECTS

German Day-Ahead Electricity Price Forecaster

Independent Project

Python LightGBM Pandas NumPy FastAPI Streamlit Docker Cloud Deployment

- Built an end-to-end ML pipeline on real German electricity market data (8,664 hourly records, SMARD API), engineering renewable-share, lag, and calendar features to model the relationship between solar and wind generation levels and day-ahead price dynamics.
- Analysed structural patterns in the data including negative-price events driven by high renewable generation, reducing forecast MAE by 62% over a seasonal naive baseline (EUR 34.00 to EUR 13.01/MWh); deployed via FastAPI and Streamlit at de-electricity-forecaster.streamlit.app.
- Documented full methodology, feature rationale, and evaluation results for reproducibility and further development.

CNNs vs. Transformers: Sentiment Analysis on SST-2

Universität Bielefeld

Python PyTorch TensorFlow BERT Hugging Face Transformers NLP Deep Learning

- Fine-tuned BERT-base with progressive layer freezing on 67,349 labelled SST-2 fragments achieving 90.14% validation accuracy and 0.9014 weighted F1; implemented systematic CNN vs. Transformer benchmarking to evaluate scalability and efficiency trade-offs across architectures.
- Implemented the full experiment pipeline in PyTorch and TensorFlow: tokenisation, training loops, metric logging, and reproducible result reporting for rigorous architecture comparison.

Career OS - AI Agent Platform

Independent Project, in progress

Python FastAPI Groq LLM (llama-3.x) RAG Qdrant PostgreSQL Docker AWS ECS GitHub Actions

- Developed and deployed an end-to-end AI prototype: multi-agent platform with LLM integration, RAG pipeline over Qdrant vector storage with grounded output validation, and full production deployment to AWS ECS via Docker and GitHub Actions CI/CD.
- Researched and evaluated current LLM models for distinct task requirements; engineered prompt templates per agent function with systematic failure-mode testing.

Cardiovascular Disease Risk Prediction

Independent Project

Python Scikit-learn XGBoost Random Forest Logistic Regression Pandas NumPy

- Processed and validated 308,854 CDC BRFSS 2022 survey records, resolving an 11.4:1 class imbalance; benchmarked four classification models (AUC-ROC 0.836, F1 0.611) with structured evaluation and stakeholder-ready reporting.
- Applied Random Forest feature importance to identify dominant predictors; documented all evaluation decisions and metric rationale for transparent, reproducible reporting.

SKILLS

Programming:	Python, SQL, PostgreSQL, MySQL, Google BigQuery
ML & Deep Learning:	Machine Learning, Deep Learning, NLP, BERT Fine-tuning, PyTorch, Tensor-Flow, Scikit-learn, LightGBM, XGBoost, Hugging Face Transformers, Feature Engineering, Time-Series Forecasting
Data Engineering:	ETL Pipelines, API Integration, Data Quality, Medallion Warehouse (dbt), FastAPI, Streamlit, Plotly, Docker, Git, GitHub, Jupyter
AI & GenAI:	LLM Integration, RAG Pipelines, Prompt Engineering, Groq API, llama-3.x, Qdrant, pgvector
Infrastructure:	AWS ECS, GitHub Actions CI/CD, Terraform, Celery, Redis, Prefect
Currently Learning:	Computer Vision (OpenCV), Microsoft Azure (AZ-900)

LANGUAGES

English	Advanced (C2)	German	Beginner (A2), actively studying toward B1	Hindi	Native
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CERTIFICATIONS

- Google AI Essentials Specialization Google / Coursera (Jul 2026)
- Data Scientist Professional with Python DataCamp